

Homework 8

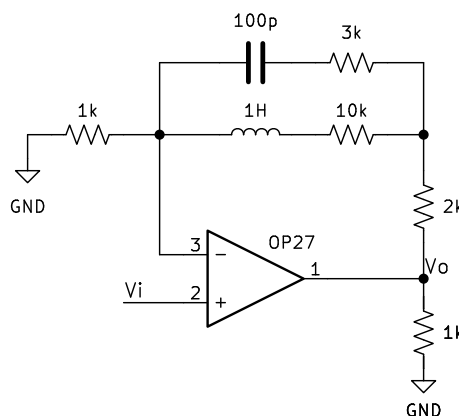
Due March 29th

Problems:

1. An op amp circuit is built to have a return ratio as given below where β is purely resistive but not necessarily known.

$$T(jf) = \frac{10^5}{(1 + jf/5)(1 + jf/10^5)(1 + jf/10^6)}$$

- (a) What is the gain margin for this circuit? Do not estimate.
 - (b) What is the phase margin for this circuit? Do not estimate.
 - (c) How much should β be increased or decreased (multiplied by or divided by) to get a phase margin of 45° ?
 - (d) How much could β be increased or decreased (multiplied by or divided by) before the circuit becomes unstable?
2. An amplifier is being designed that has a DC gain of 200 kV/V, and two high-frequency poles at 100 kHz and 50 MHz. Note that in its current design, it is *not* a constant gain bandwidth product amplifier.
 - (a) What is the maximum value for β such that no gain peaking occurs? Assuming the system corresponding circuit is a non-inverting amplifier, what is the minimum closed loop gain?
 - (b) A capacitor is to be added within the amplifier design (for example, see Book Figure 5.1) to add a low frequency pole in the amplifier open loop response. At what maximum frequency does this pole need to be placed for $\beta = 1$ to result in $\phi_M = 60^\circ$?
 3. Recreate this circuit in LTspice and measure its $T(jf)$ **in two different locations** via the Middlebrook technique. For each location, perform both voltage and current injection and determine if one or both are needed to determine $T(jf)$ and explain why. Perform the characterization of $T(jf)$ over the frequency range of 1 Hz to 10 MHz/ Verify that the resultant $T(jf)$ is functionally equivalent between the two test points (excluding possible simulation inaccuracies, similar to what was shown in lecture). The amplifier is powered by $\pm 5V$ supplies. Provide a screenshot of the built circuit showing the use of the Middlebrook technique at each point and any applicable output plots.



4. Consider a two-pole non-GBP amplifier with the open loop gain given below, use Rate of Closure analysis and the axes below to answer each subquestion.

$$a(jf) = \frac{10^5}{(1 + jf/10)(1 + jf/10^4)}$$

- Begin by plotting the straight-line estimated representation for $|a(jf)|$. Label the axes.
- Estimate the cross-over frequency, f_x , and phase margin for each of the three constant β values: $\beta = 0.1$, $\beta = 1/10^2$, $\beta = 1/10^4$. Label each corresponding line on the graph.
- Assume β also contains a pole, such that $\beta = 1/(10^3(1 + jf/f_p))$. What is the minimum value for f_p such that the circuit is stable?

